

Nils Lukas

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Research Interests	Design secure and private AI agents in the presence of untrustworthy providers , data , models , and users . 1. Providers: Confidential computing & encrypted inference. 2. Data: Poisoning, prompt injection & training-data privacy. 3. Models: Scalable oversight & differentially private inference. 4. Users: Detect misuse via watermarking & safety monitoring.
Education	University of Waterloo, Canada2019 – 02/2024 Ph.D. in Computer Science • Advisor: Florian Kerschbaum • Thesis: Analyzing Threats of Large-Scale Machine Learning Systems • Thesis Awards: Top Mathematics Doctoral Prize & Alumni Gold Medal RWTH-Aachen, Germany M.Sc. in Computer Science (<i>w/Distinction</i>)2016 – 2018 B.Sc. in Computer Science10/2012 – 2016
Honors & Awards	Amazon Research Award [100,000 USD]2025 First Place at the NeurIPS'24 Watermarking Competition [4,400 USD]2024 First Place at DGE Elite Hackathon, GITEX'24 [10,900 USD]2024 Top Mathematics Doctoral Thesis, University of Waterloo [1,080 USD]2024 Alumni Gold Medal, One PhD Award Yearly, University of Waterloo2024 Best Poster Award, Sponsored by David R. Cheriton [220 USD]2023 Distinguished Contribution Award, Microsoft MLADS conference2023 David R. Cheriton Scholarship, University of Waterloo [14,400 USD]2022, 2023 Outstanding Reviewer, ICML'222022 Best Poster Award, Sponsored by Rogers [720 USD]2019
Research Grants	Awarded\$1,458,000 [United AI-Saqer Group] MedBoard: Privacy-preserving Multi-Agent AI for Healthcare PI: Nils Lukas · Co-PIs: Praneeth Vepakomma, Preslav Nakov \$136,000 · 2026 [IAAI Research Program] Trust Layer for Safe Smallholder Advisory PI: Salem Lahlou · Co-PIs: Nils Lukas, Martin Takáč \$500,000 · 2026 [Amazon Research Awards – Special Call] AdvSim2Real: Robustifying Agents in Simulated Adversarial Environments PI: Nils Lukas · Co-PI: Praneeth Vepakomma \$100,000 · 2025 [Etihad Airways] Conversational Booking Agents PI: Nils Lukas · Co-PIs: Salem Lahlou, Alham Fikri Aji, Martin Takáč, Mingming Gong \$450,000 · 2025 [United AI-Saqer Group] Privacy-preserving Brain–Computer Interfaces PI: Abdulrahman Mahmoud · Co-PIs: Nils Lukas, Elizabeth Churchill \$136,000 · 2025 [TII Funding] GFlowNets for Fuzzing of Agentic Applications PIs: Salem Lahlou & Nils Lukas \$136,000 · 2025
Conference Publications	<u>Underlined</u> authors are students and postdocs I supervise as primary advisor.

23	SPQR: A Multi-Dimensional Benchmark for Safety Alignment under Benign Model Adaptation Mohammed Talha Alam, Nada Saadi, Fahad Shamshad, Nils Lukas , Karthik Nandakumar, Fakhri Karray, <u>Samuele Poppi</u> . European Conference on Computer Vision, 2026.
22	RAVEN: Erasing Invisible Watermarks via Novel View Synthesis Fahad Shamshad, Nils Lukas , Karthik Nandakumar. IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.
[CVPR'26] AR: 25.4% (4,090/16,092) 🏆 Oral (top 0.8%)	
21	Collaborative Threshold Watermarking <u>Tameem Bakr</u> , Anish Ambreth, Nils Lukas . The Forty-Third International Conference on Machine Learning, 2026.
[ICML'26] AR: 26.6% (6,352/23,918)	
20	CoRe: Collaborative Reasoning via Cross Teaching <u>Kshitij Mishra</u> , Mirat Aubakirov, Martin Takáč, Nils Lukas , Salem Lahlou. The Forty-Third International Conference on Machine Learning, 2026.
[ICML'26] AR: 26.6% (6,352/23,918)	
19	Forest Before Trees: Latent Superposition for Efficient Visual Reasoning Yubo Wang, Juntian Zhang, Yichen Wu, Yankai Lin, Nils Lukas , <u>Yuhan Liu</u> . The 64th Annual Meeting of the Association for Computational Linguistics, 2026.
[ACL'26] Main Conference AR: 18.9% (2,296/12,148)	
18	Detecting AI-Generated Video: A Vision-Language Dual-View Survey Dylan Xinming Hou, Juntian Zhang, Xu Gu, Yichen Wu, Nils Lukas , Gus Xia, Xiuying Chen, <u>Yuhan Liu</u> . Findings of the Association for Computational Linguistics, 2026.
[ACL'26] Findings AR: 17.8% (2,163/12,148)	
17	DP-Fusion: Token-Level Differentially Private Inference for Large Language Models <u>Rushil Thareja</u> , Preslav Nakov, <u>Praneeth Vepakomma</u> , Nils Lukas . The Fourteenth International Conference on Learning Representations, 2026.
[ICLR'26] AR: 27.4% (5,355/19,525)	
16	SD-E2: Semantic Exploration for Reasoning Under Token Budgets <u>Kshitij Mishra</u> , Nils Lukas , Salem Lahlou. Findings of the European Chapter of the Association for Computational Linguistics, 2026.
[EACL'26] Findings	
15	Mask Image Watermarking Runyi Hu, Jie Zhang, Shiqian Zhao, Nils Lukas , Jiwei Li, Qing Guo, Han Qiu, Tianwei Zhang. The Thirty-Ninth Annual Conference on Neural Information Processing Systems, 2025.
[NeurIPS'25] AR: 24.5% (5,290/21,575)	
14	SPIRIT: Patching Speech Language Models against Jailbreak Attacks Amir Djanibekov, Nurdaulet Mukhituly, Kentaro Inui, Hanan Aldarmaki, Nils Lukas . Empirical Methods in Natural Language Processing (Main Conference), 2025.
[EMNLP'25] AR: 22.2% (1,811/8,172)	
13	Optimizing Adaptive Attacks against Content Watermarks for Language Models Abdulrahman Diao, <u>Toluwani Aremu</u> , Nils Lukas . The Forty-Second International Conference on Machine Learning, 2025.
[ICML'25] AR: 26.9% (3,260/12,107) ★ Spotlight (Top 2.6%)	
12	Cowpox: Towards the Immunity of VLM-based Multi-Agent Systems Yutong Wu, Jie Zhang, Yiming Li, Chao Zhang, Qing Guo, Han Qiu, Nils Lukas , Tianwei Zhang. The Forty-Second International Conference on Machine Learning, 2025.
[ICML'25] AR: 26.9% (3,260/12,107)	

- 11
[USENIX'24]
AR: 19.1%
(417/2,176)

PEPSI: Practically Efficient Private Set Intersection in the Unbalanced Setting
Rasoul Mahdavi, **Nils Lukas**, Faezeh Ebrahimiaghazani, Thomas Humphries, Bailey Kacsmar, John Premkumar, Xinda Li, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.
- 10
[USENIX'24]
AR: 19.1%
(417/2,176)

Fast and Private Inference of Deep Neural Networks by Co-designing Activation Functions
Abdulrahman Diaa, Lucas Fenaux, Thomas Humphries, Marian Dietz, Faezeh Ebrahimiaghazani, Bailey Kacsmar, Xinda Li, **Nils Lukas**, Rasoul Akhavan Mahdavi, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.
- 9
[ICLR'24]
AR: 30.8%
(2,250/7,262)

Leveraging Optimization for Adaptive Attacks on Image Watermarks
Nils Lukas, Abdulrahman Diaa, Lucas Fenaux, Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.
- 8
[ICLR'24]
AR: 30.8%
(2,250/7,262)
● Media Coverage

Universal Backdoor Attacks
Benjamin Schneider, **Nils Lukas**, Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.
- 7
[USENIX'23]
AR: 29.2%
(422/1,444)

PTW: Pivotal Tuning Watermarking for Pre-Trained Image Generators
Nils Lukas and Florian Kerschbaum. In the 32nd USENIX Security Symposium, 2023.
- 6
[S&P'23]
AR: 17.0%
(195/1,147)
★ Distinguished Contribution Award at Microsoft MLADS

Analyzing Leakage of Personally Identifiable Information in Language Models
Nils Lukas, Ahmed Salem, Robert Sim, Shruti Tople, Lukas Wutschitz, Santiago Zanella-Béguelin. In the 44th IEEE Symposium on Security and Privacy, 2023.
- 5
[S&P'22]
AR: 14.5%
(147/1,012)

SoK: How Robust is Image Classification Deep Neural Network Watermarking?
Nils Lukas, Edward Jiang, Xinda Li, Florian Kerschbaum. In the 43rd IEEE Symposium on Security and Privacy, 2022.
- 4
[ICLR'21]
AR: 28.7%
(860/2,997)
★ Spotlight (Top 5%)

Deep Neural Network Fingerprinting by Conferrable Adversarial Examples
Nils Lukas, Yuxuan Zhang, Florian Kerschbaum. The Ninth International Conference on Learning Representations, 2021.
- 3
[IH&MMSEC'21]
AR: 40.3% (128/318)

On the Robustness of Backdoor-based Watermarking in Deep Neural Networks
Masoumeh Shafieinejad, **Nils Lukas**, Jiaqi Wang, Xinda Li, Florian Kerschbaum. Proceedings of the 2021 ACM Workshop on Information Hiding and Multimedia Security, 2021.
- 2
[ACSAC'20]
AR: 20.9% (104/497)

Practical Over-Threshold Multi-Party Private Set Intersection
Rasoul Mahdavi, Thomas Humphries, Bailey Kacsmar, Simeon Krastnikov, **Nils Lukas**, John Premkumar, Masoumeh Shafieinejad, Simon Oya, Florian Kerschbaum, Erik-Oliver Blass. Annual Computer Security Applications Conference (ACSAC), 2020.

	1	Differentially Private Two-Party Set Operations	
	[EuroS&P'20] AR: 20.9% (39/187)	Bailey Kacsmar, Basit Khurram, Nils Lukas , Alexander Norton, Masoumeh Shafieinejad, Zhiwei Shang, Yaser Baseri, Maryam Sepehri, Simon Oya, Florian Kerschbaum. IEEE European Symposium on Security and Privacy (EuroS&P), 2020.	
Journal Publications	2	Privacy-Preserving Machine Learning [Cryptography]	
	[IEEE S&P'23]	Florian Kerschbaum, Nils Lukas . IEEE Security & Privacy, Volume 21, Issue 6, 2023.	
Workshop Papers	1	SunFlower: A new Solar Tower Simulation Method for use in Field Layout Optimization	
	[AIP'18]	Pascal Richter, Gregor Heiming, Nils Lukas , Martin Frank. AIP Conference Proceedings, Volume 2033, Issue 1, 2018.	
	3	Sanitizing Medical Documents with Differential Privacy using Large Language Models	
	[GenAI4Health]	Rushil Thareja, Gautam Gupta, Preslav Nakov, Praneeth Vepakomma, Nils Lukas . 2025.	
Working Papers	2	First-Place Solution to NeurIPS 2024 Invisible Watermark Removal Challenge	
	[WMARK'25]	Fahad Shamshad, Tameem Bakr, Yahia Salaheldin Shaaban, Noor Hazim Hussein, Karthik Nandakumar and Nils Lukas . The 1st Workshop on GenAI Watermarking, 2025.	
	1	Optimizing Adaptive Attacks against Content Watermarks for Language Models	
	[WMARK'25] ★ Oral Presentation	Abdulrahman Diaa, Toluwani Aremu, Nils Lukas . The 1st Workshop on GenAI Watermarking, 2025.	
Research Talks		Mitigating Watermark Forgery in Generative Models via Randomized Key Selection , Submitted.	
		Toluwani Aremu, Noor Hazim Hussein, Munachiso S Nwadike, Samuele Poppi, Jie Zhang, Karthik Nandakumar, Neil Zhenqiang Gong, Nils Lukas . 2025.	
		Robust and Calibrated Detection of Authentic Multimedia Content , Submitted.	
		Sarim Hashmi, Abdelrahman Elsayed, Mohammed Talha Alam, Samuele Poppi, Nils Lukas . 2025.	
Research Talks	AI Safety Workshop: Alignment, Oversight and Deception		
	•	ADIA Lab & UGR Summer School , Granada	2026
	The Role of Watermarking for ML Security		
	•	CyberQ , UAE Cyber Security Council & TII, Abu Dhabi	2025
	Adaptively Robust and Forgery-Resistant Watermarking		
	•	Meta (FAIR), hosted by Hady Elsahar	2025
	•	International Symposium on Trustworthy Foundation Models, MBZUAI	2025
	Emerging Topics in Machine Learning		
	•	Build with AI , Google Developer Groups, Abu Dhabi	2025
	Optimizing Adaptive Attacks against Content Watermarks		
	•	DeepMind, hosted by David Stutz	2024
	•	University of California, Berkeley, hosted by Dawn Song	2024
	Analyzing Leakage of Personal Information in Language Models		
	•	Microsoft M365, hosted by Robert Sim	2024
	•	Meta, hosted by Will Bullock	2023
	•	MongoDB, hosted by Marilyn George and Archita Agarwal	2023

	How Reliable is Watermarking for Image Generators?	
	• Google, hosted by Somesh Jha	2023
	• University of California, Berkeley, hosted by Dawn Song	2023
Keynotes	Aviation Future Week , hosted by Emirates, Dubai	2024
	Cyber Energy Leadership Forum , Abu Dhabi	2024
Work Experience	Assistant Professor , MBZUAI, Abu Dhabi, UAE	since 08/2024
	Research Intern , Royal Bank of Canada, Borealis AI, Toronto	2024
	• Vertical Federated Learning, hosted by Kevin Wilson	
	Research Intern , Microsoft Research, Cambridge, UK	2022
	• Privacy for Language Models, hosted by Shruti Tople & Lukas Wutschitz	
	Research Assistant , RWTH-Aachen, Aachen	2014 – 2018
	Student Researcher , DSA Daten- und Systemtechnik GmbH, Aachen	2016
	Software Engineer Intern , A.R. Bayer DSP Systeme GmbH, Düsseldorf	2012
Teaching	Instructor, MBZUAI, UAE	
	• ML8509: Collaborative Machine Learning (w/ S. Horvath)	2026
	• ML8502: Machine Learning Security (14 weeks)	2025
	• ML807: Federated Learning (7 weeks)	2025
	• ML818: Emerging Topics in Trustworthy Machine Learning (4 weeks)	2024
	Teaching Assistant, University of Waterloo, Canada	
	• CS458/658: Computer Security and Privacy	2020, 2021
	• CS246 – Object Oriented Programming	2021
	Co-Instructor, RWTH-Aachen, Germany	
	• Course: Data-driven Medicine	2018
Service	Area Chair	
	• AAAI Conference on Artificial Intelligence (AAAI), Senior Program Committee	2027
	• Neural Information Processing Systems (NeurIPS)	2026
	• International Conference on Learning Representations (ICLR)	2026
	• International Conference on Machine Learning (ICML)	2026
	Program Committee	
	• ACM Conference on Computer and Communications Security (CCS)	2025
	• IEEE Symposium on Security and Privacy (IEEE S&P)	2025–2027
	• Recent Advances in Intrusion Detection (RAID)	2024
	Artifact Evaluation Committee	
	• The ACM Conference on Computer and Communications Security (CCS)	2023, 2024
	Reviewer	
	• European Conference on Computer Vision (ECCV)	2026
	• Conference on Language Modeling (COLM)	2026
	• British Machine Vision Conference (BMVC)	2026
	• FAGEN Workshop @ ICML	2026
	• Workshop on GenAI Watermarking (WMark @ ICLR)	2025
	• NETYS	2025
	• ACM TheWebConf (WWW)	2025
	• International Conference on Learning Representations (ICLR)	2024, 2025
	• International World Wide Web Conference (TheWebConf)	2024
	• Recent Advances in Intrusion Detection (RAID)	2023
	• Neural Information Processing Systems (NeurIPS)	2022, 2023
	• International Conference on Machine Learning (ICML)	2022, 2025

- The Conference on Information and Knowledge Management (CIKM) 2020

Journal Reviewer

- Communications of the ACM (CACM) 2025

Other

- External Examiner, PhD Dissertation, CISPA Helmholtz Center for Information Security 2026
- Faculty Search Committee, Machine Learning Department at MBZUAI 2025–2026
- Chair for the invited faculty talk program, International Symposium on Trustworthy Foundation Models at MBZUAI 2025
- Admission's Committee, MBZUAI Machine Learning Department 2025
- Student Board Member, Cybersecurity and Privacy Institute 2022–2024
- Session Chair, IEEE Symposium on Security and Privacy (S&P) 2023
- Sub-Reviewer, Proceedings on Privacy Enhancing Technologies (PETS) 2021–2023
- School Advisory Committee on Appointments Liaison, CrySP Lab 2022
- Organization, Workshop on Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA) 2018

PostDocs, Students

Postdoctoral Researchers

- [Samuele Poppi](#) since 2025
- [Kshitij Mishra](#) (*secondary: [Martin Takáč](#) & [Salem Lahlou](#)*) since 2025
- [Dushyant Chauhan](#) (*secondary: [Martin Takáč](#) & [Salem Lahlou](#)*) since 2025
- [Yuhan Liu](#) (*secondary: [Martin Takáč](#) & [Salem Lahlou](#)*) since 2025

Research Engineers

- [Sarim Hashmi](#) since 2025

PhD Students

- [Fabian Zhang](#) (ML) since 2025
- [Tair Djanibekov](#) (NLP) since 2025
- [Rushil Thareja](#) (NLP) (*secondary: [Praneeth Vepakomma](#)*) since 2024
- [Toluwani Aremu](#) (ML) since 2023

MSc Students

- [Juan Nicolas Sepulveda Arias](#) (CV) since 2025
- [Mohamed Mahmoud Mohamed Hendy](#) (ML) since 2025
- [Majid Mohamed Mohamed Sulaiman Ibrahim](#) (ML) since 2025
- [Prince Jha](#) (ML) (*primary: [Salem Lahlou](#)*) since 2025
- [Muhammad Mohideen](#) (CV) (*secondary: [Samuel Horvath](#)*) since 2025

Alumni

- [Tameem Bakr](#) (ML) MSc · Mar 2026
- [Shaikha Jasem Mohamed Ismail Alhosani](#) (ML) MSc · Mar 2026
- [Maryam Abdulla Rashed Abdulla Alshamsi](#) (ML) MSc · Mar 2026
- [Maryam Alshehyari](#) (ML) MSc · Mar 2026
- [Abdalla Khalid Mohamed Abdalla Alzaabi](#) (ML) MSc · Mar 2026
- [Ali Al-Ali](#) (ML) MSc · Mar 2026